Basics of Programming

Semester	Course Code	Name of the course	L	T/A	P	Credits
B.Tech. First Year /Sem I / Sem-II		Basics of Programming	3	0	0	3

Prerequisites for the course	
1	Basics of Programming and Logic such as operators, keywords, flowcharts

Prior Reading Material/ useful links	
1	C Programming Absolute Beginner's Guide (3rd Edition)' by Greg Perry and Dean
2	Millerhttps://www.digimat.in/nptel/courses/video/105107176/L01.html
3	https://www.youtube.com/watch?v=8mHTieXk1x8

Course Outcomes:

Sr. No.	Course outcome number	CO statement
1	CO 1	Understand basic Structure of the C-PROGRAMMING, declaration and usage of variables.
2	CO 2	Demonstrate an understanding of algorithms in the problem-solving process.
3	CO 3	Understand the arrays and strings and create programs using them.
4	CO 4	To Understand the basics of oops concepts.
5	CO 5	Make use of different data structure like Linear and Non-Linear.

Syllabus:

Course Contents	
Unit I	**Overview of C Programming:** Introduction to C: History of 'C', General structure of a 'C' program, Flowchart, Algorithm, Header files, 'main' function. Data Concepts: Character set, tokens, keywords, Identifiers, Variables, Constants, data types, C operators, Arithmetic operators, Relational operators, Logical operators, Assignment operators, Bitwise operators, Increment/Decrement operators, Conditional operators and data type conversion. Basic Input output: Input and Output statements, using printf( ) and scanf( ), Formatted Input/Output statements. Basic structure of C program, Executing a 'C' program **[6 Hrs]**
Unit II	**Control Structures:** Decision making and Branching: if statements, if else statements, nested if else statements, if-else ladder, The Switch statements. Looping: While loop, Do While loop, for loop, Go to statements, Use of break and continue statements **[6 Hrs]**
Unit III	**Array, Structure and Function:** Arrays: One dimensional array, two-dimensional array, array declaration and initialization, accessing array. Array of characters, operation on array.

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	Introduction and Features of structures: Defining the structures, Declaring and Accessing Members of structures, Initialization of Structures, Array of Structures, Concept and need of functions: User defined functions, function declaration, function call, Library functions: Math functions, string handling functions, Storage Classes, Concept of Pointers: Declaring, initializing, accessing pointers. **[6 Hrs]**
Unit IV	**Basics of Object-Oriented Programming:** Introduction to C++, OOPs Concepts: Class & Object, Inheritance: Types of Inheritance, base class, derived class, Polymorphism, Encapsulation, Abstraction, Access Specifiers, Data types in C++, Structure of C++ Program, The standard Input/Output functions. **[6 Hrs]**
Unit V	**Introduction to Data Structures:** Concepts and need of Data Structure, Data Types- Primitive and Non-Primitive, Abstract Data Types, Types of Data Structures- Linear and Non-Linear Data structure, basic concept of Stack, Queue, Linked List, Tree, Graph, Sorting and Searching, Types of Sorting and Searching. **[6 Hrs]**

	Text Books
1	Let Us C solutions by Yashavant Kanetkar
2	Reema Thareja, - Data Structures Using C, Second Edition , Oxford University Press, 2011

	Reference Books
1	C Programming: A Modern Approach (2nd Edition) - K. N. King (2008). A good book for learning C..
2	Programming in C (4th Edition) - Stephen Kochan (2014). A good general introduction and tutorial
3	Ilis Horowitz, Sartaj Sahni, Susan Anderson-Freed, —Fundamentals of Data Structures in C, Second Edition, University Press, 2008.

	Useful links
1	https://www.unf.edu/~wkloster/2220/ppts/cprogramming_tutorial.pdf
2	http://www.lmpt.univ-tours.fr/~volkov/C++.pdf

Contributions for syllabus designing:

Sr. No.	Name of the person	Designation	Organization
1	Mr. Yuvraj Suryavanshi	Assistant Professor	JDCOEM
2	Ms. Raksha Swami	Alumni	JDCOEM

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